

Five Maximal Incoming Classes Suffice: Exact Certificates and Structural Constraints for Seymour's Second-Neighbourhood Conjecture

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Abstract

We study Seymour's second-neighbourhood conjecture through incoming-neighbourhood patterns, counterexample-preserving row maps, and protection-preserving tournament completions. A finite system of homogeneous pattern inequalities provides an exact-second-neighbourhood certificate in the original graph. We prove that universal feasibility of these systems is equivalent to the conjecture, and verify feasibility for every oriented core on at most five vertices using nonnegative integer certificates. Consequently, every finite nonempty oriented graph with at most five inclusion-maximal incoming-neighbourhood classes has a whole maximal class of Seymour vertices, with no bound on its total order. We also prove a simultaneous outgoing-row copying lemma, obtain an independent-twin normal form for a lexicographically extremal counterexample, and locate a global core of gap-one maximal incoming classes. Iterated protection closure preserves the original first and exact second neighbourhoods at its partial median feeds. The resulting tournament feeds have a strict original far-head surplus; at zero completed gap, a safe weak closure forces an original cycle among the remaining missing partners. In the arc-set-minimal setting this gives residual at least four at every such zero-gap feed. Together with the supplementary residual exclusions, we obtain the necessary bound $2M - n \geq 7$, strengthened to eight at even order, where n is the order and M is the number of missing unordered vertex pairs in the arc-set-minimal counterexample under consideration. All finite certificates and independent integer verifiers are supplied. These results do not prove or refute the conjecture in full generality.

Keywords: oriented graph; second neighbourhood; tournament; median order; incoming-neighbourhood inclusion; computer-assisted proof. **MSC 2020:** 05C20, 05C22, 05C85.

1 Introduction

For a vertex v of a finite oriented graph D , let $N_D^+(v)$ denote its outneighbourhood and let $N_D^{++}(v)$ consist of vertices whose directed distance from v is exactly two. Seymour's conjecture asks whether every nonempty oriented graph has a vertex satisfying

$$|N_D^{++}(v)| \geq |N_D^+(v)|. \quad (1.1)$$

Such a vertex is called a Seymour vertex. The exclusion of first neighbours from the second set is essential in every deletion, completion and counting argument below.

The tournament case is known, but a general completion may create second neighbours that were absent in the original graph. We develop two ways of making that discrepancy explicit. The first uses a small induced incoming core and an exact linear certificate. The second completes only carefully controlled directions before a global median-order optimization. A separate copying operation imposes global structural restrictions on a suitably chosen counterexample. The three constructions have different preservation properties; their hypotheses are kept distinct throughout the paper.

1.1 Principal results

Partition vertices by equality of their full incoming neighbourhoods. A class is maximal when its common incoming set is maximal under inclusion among the sets $N_D^-(v)$. Maximality refers to inclusion, not to the numerical indegree. Our first unconditional result is the following computer-assisted theorem.

Theorem 1.1 (Five maximal incoming classes). *Every finite nonempty oriented graph with at most five maximal incoming-neighbourhood classes has a maximal class all of whose vertices are Seymour vertices. The total order and the sizes of the classes are unrestricted.*

The theorem follows from a transfer inequality valid for arbitrary induced cores. Its finite component comprises 59 809 labelled cores of orders one through five and 462 404 nonempty pattern inequalities. Every witness is a nonnegative integer vector, with entries at most three. The tables are checked independently of the program that found them. The full proof and the precise finite verification contract appear in Theorem 3.8.

The same certificate system has a converse: an infeasible core system yields, by a finite labelled construction, a genuine all-deficient oriented graph (Theorem 3.10). Thus universal feasibility is equivalent to the full conjecture. A certificate also lifts through repeated maximal incoming-core reductions, reducing the remaining feasibility question to cores with pairwise distinct, incomparable incoming sets.

Our second group of results concerns simultaneous copying. If an original vertex map ρ satisfies $N_D^-(v) \subseteq N_D^-(\rho(v))$ for every v , replacing each outgoing row by the original row of $\rho(v)$ creates an oriented graph whose gap at v is at least the original gap at $\rho(v)$. This implies a global covering property in a smallest-order counterexample: every vertex receives an entire maximal incoming class whose members all have gap one. A further extremal selection makes equal incoming neighbourhoods into independent true-twin modules. These statements use global extremality rather than merely inclusion-minimal deletion assumptions.

For the completion results write $g_D(v) = d_D^+(v) - d_D^{++}(v)$ and let $t_D(v)$ count the original far vertices not protected by incoming-neighbourhood nesting. The closure constructed in Section 5 admits an optimal reference completion whose every original-positive-gap median feed satisfies

$$t_D(f) \geq g_D(f) + 1. \quad (1.2)$$

Under the safe weak extension, zero completed gap requires at least three remaining missing partners, an original directed cycle among them, and $t_D(f) \geq g_D(f) + 3$. For a strongly connected all-deficient graph minimal under every nonempty arc-set deletion, the same feed satisfies $\eta_D(f) \geq 4$, where $\eta_D(v) = \mu_D(v) - 1 + d_D^-(v) - d_D^{++}(v)$.

The supplementary case reductions, together with an unbounded one-exception theorem for actual bad-direction failure sources, yield

$$2M - n \geq 7, \quad M \geq \lceil n/2 \rceil + 3 + \mathbf{1}_{\{n \text{ even}\}}. \quad (1.3)$$

The missing-pair bound is a necessary condition for the stated arc-set-minimal class. It is not a missing-edge density theorem for arbitrary original graphs: deleting arcs changes M . Proofs of the residual allocations and the specialized tools for $n = 2\delta^+ + 3$ are collected in the companion supplement.

1.2 Relation to previous work

The tournament case of Seymour’s second neighborhood conjecture was proved by Fisher [7] using a probability distribution obtained from a skew-symmetric game. Havet and Thomassé [12] subsequently gave a proof using median orders, including a stronger bound involving the good vertices of an order. Both methods enter the present work. The incoming-core argument uses the existence of a losing density on a tournament; the completion arguments use the strong feed bound and the equality case of sedimentation. These are established inputs. Our arguments specify the additional conditions needed to transfer information from a tournament completion to the exact second neighborhoods of the original graph. In particular, the existence of a Seymour vertex in a completion alone does not provide such a transfer.

Fidler and Yuster [6] proved the vertex-weighted tournament statement and established further cases obtained by deleting prescribed edge sets from tournaments. Ghazal’s work on generalized stars and other missing graphs [8, 9] develops the convenient orientation and good-missing-edge approach underlying several of our completion definitions. Dara, Francis, Jacob and Narayanan [4] extended the matching case to a missing graph whose edges can be partitioned into a matching and a star, and also considered partitions into an independent set and a 2-degenerate graph. Our protection-preserving completions require additional compatibility conditions beyond convenience of individual added arcs. The proof of each transfer inequality therefore accounts for paths using added arcs and for original second neighbors that become first neighbors. None of these inequalities is inferred simply by applying a known tournament theorem to an arbitrary completion.

Minimal-counterexample methods provide another starting point. Brantner, Brockman, Kay and Snively [2] established strong connectivity, gaps in $\{1, 2\}$, and restrictions involving gap-one predecessors and directed cycles for their minimal counterexamples. Their extremal selection first minimizes the number of arcs over all counterexamples and then minimizes the order. The selections used here must be distinguished from that convention: fixed-order minimality under deletion of a nonempty arc set, and minimum order followed by arc count and further finite objectives, serve different arguments. We give the required deletion and copying arguments for the stated selection in each case. In particular, the predecessor-containment row operation and the resulting normal form are not treated as consequences of the earlier extremal definition.

Seacrest’s subset formulation [16] supplies a closely related boundary perspective. Its edge-minimal subset inequality motivates the boundary budgets used in the residual analysis. Because the sets and minimality assumptions have to agree exactly, the external-boundary form needed here is proved explicitly. This paper is distinct from Seacrest’s arc-weighted extension [15]; an arc-weighted result is not substituted for the unweighted feed or boundary statements. The residual bound $R \geq 7$ in this manuscript concerns the specified minimal all-deficient graphs. It must not be read as an assertion about the missing-edge count of every counterexample before a minimizing deletion.

There are further reasons to keep extremal assumptions visible. Espuny Díaz, Girão, Granet and Kronenberg [5] prove that a vertex-minimal counterexample has minimum out-

degree greater than the square root of its order, while also constructing arbitrarily large strongly connected counterexamples with bounded minimum outdegree, conditional on the existence of any counterexample. Zelenskiy, Damosiuk and Nalivayko [18] obtain conditional counterexample constructions with widely differing densities and diameters. Thus counterexample amplification, and cyclic substitution in particular, already have a literature. The calculations here involving such constructions are used to delimit particular completion or weighted transfer claims, rather than to assert a new general amplification principle.

Theorem 3.8 has a different parameter from these density and degree results: the number of distinct inclusion-maximal incoming neighborhoods. There is no restriction on the order of the original graph or on the size of an incoming-neighborhood class. The certificate on a chosen core is checked against every permitted subset of an incoming set, so that it controls all original target patterns simultaneously. An ordinary verification of Seymour’s conjecture for small graphs does not automatically supply these stronger certificates. The finite tables are therefore accompanied by explicit integer checks and a proof of their transfer to arbitrary-order graphs. The labelled construction from an infeasible pattern system establishes a converse at the universal level. Finite linear-programming duality is standard; the graph realization and its exact-distance calculation are the substantive steps that must be verified for this particular reformulation. Universal feasibility remains unproved.

Recent special cases provide useful comparisons. Wang and Lu [17] prove the conjecture for $\mathcal{D}_{0,3} \cup \mathcal{D}_{1,1}$, where $\mathcal{D}_{s,t}$ consists of oriented graphs admitting a vertex partition whose induced underlying graphs are respectively s - and t -degenerate. Bai, Li and Park [1] study the stronger requirement of a complete directed matching from the first to the exact second neighborhood, proving it when the minimum outdegree is at most five or the graph is 5-anti-transitive. Kaneko and Locke [13] established ordinary Seymour vertices when the minimum outdegree is at most six. The preprint of Sadhukhan, Sandeep and Sen [14] treats minimum outdegree seven using computer-assisted local reductions. Brukhman’s preprint [3] treats the dense range $n \leq 2\delta^+ + 2$. The incoming-core theorem uses none of these recent preprints as a premise.

The five-class hypothesis does not imply the hypotheses of those particular recent results. For example, replace each vertex of the regular five-vertex tournament, with arcs $i \rightarrow i + 1, i + 2$ modulo five, by an independent set of size $m \geq 4$. The resulting graph has five maximal incoming classes, order $5m$, and minimum outdegree $2m$. Its underlying complete five-partite graph belongs to neither $\mathcal{D}_{0,3}$ nor $\mathcal{D}_{1,1}$, and a path through parts 0, 1, 2, 3, 4, 1 excludes 5-anti-transitivity. It also lies outside the stated dense range. This example itself is already covered by the weighted tournament theorem. It demonstrates noncontainment in specific hypotheses, not novelty relative to every known weighted or substitution argument.

Halkiewicz’s split-twin operation [11] duplicates incoming neighbors and partitions outgoing neighbors while increasing the order; its stated preservation theorem has a condition involving an existing Seymour vertex. This differs from the fixed-order row copying used here. Nevertheless, this distinction and the preceding comparisons do not settle priority for equivalent formulations under other terminology. The present paper makes no claim to an exhaustive classification of earlier reductions. The literature comparison is current to 7 October 2026. Public full-proof claims, including [10], are not assumed correct or used as premises. The conclusions proved here remain partial results and reformulations of the conjecture.

1.3 Organization and verification

The incoming-core and cloning arguments are independent of the residual machinery. After establishing them, we give the closure construction and the zero-gap cycle argument, then state the minimal-counterexample consequences with their supplementary proof locations. The final sections give reproducibility details, exact controls against stronger transfer claims, and the unresolved global steps. The supplement is a self-contained technical repository of the longer foundational and case arguments. It also records which earlier finite classifications are subsumed by the stronger results.

2 Notation and extremal hypotheses

All graphs are finite and nonempty. An oriented graph has no loops and no pair of oppositely directed arcs. Set

$$N_D^{++}(v) = \left(\bigcup_{a \in N_D^+(v)} N_D^+(a) \right) \setminus (N_D^+(v) \cup \{v\}), \quad g_D(v) = d_D^+(v) - d_D^{++}(v).$$

Reverse exact second neighbourhoods are denoted by $N_D^{--}(v)$. We call D all-deficient when $g_D(v) > 0$ for every vertex. A missing partner of v is a different vertex nonadjacent to it; its set and size are $\mathcal{M}_D(v)$ and $\mu_D(v)$. The number of missing unordered pairs is $M(D)$, so $\sum_v \mu_D(v) = 2M(D)$.

The far set and the reverse far set are

$$U_D(v) = V(D) \setminus (\{v\} \cup N_D^+(v) \cup N_D^{++}(v)), \quad P_D(v) = \{x \neq v : v \in U_D(x)\}.$$

For $S \subseteq V(D)$ we use external boundaries

$$\Gamma_D(S) = \bigcup_{x \in S} N_D^+(x), \quad \partial_D S = \Gamma_D(S) \setminus S, \quad \partial_D^2 S = \Gamma_D(\partial_D S) \setminus (S \cup \partial_D S).$$

The word external distinguishes these sets from conventions that allow a set neighbourhood to meet the starting set.

We will use three different extremal choices. An all-deficient graph is *arc-set-minimal* when deleting every nonempty set of arcs on its fixed vertex set destroys all-deficiency. The abbreviation MIN means, in addition, that it is strongly connected. A *minimum-order* counterexample has the smallest order among all counterexamples, including graphs obtained by rewiring. The lexicographic normal form in Section 4 adds further global objectives. These conditions are not interchangeable. If a counterexample exists, a MIN graph exists by taking an arc-minimal spanning subgraph and then a sink strong component. A minimum-order counterexample is automatically strongly connected.

Definition 2.1 (Protection). Write $a \triangleleft_D b$ if $a \rightarrow b$ and $N_D^-(a) \subseteq N_D^-(b) \setminus \{a\}$. Then a is a protected far head of b . This is a strict partial order. Put

$$\mathcal{F}_D(b) = U_D(b) \setminus \{a : a \triangleleft_D b\}, \quad t_D(b) = |\mathcal{F}_D(b)|.$$

Indeed, an alleged path $b \rightarrow x \rightarrow a$ would force $x \rightarrow b$, a digon. Transitivity follows because $a \rightarrow b \rightarrow c$ and predecessor containment give $a \rightarrow c$ and the required further containment. A whole vertex, meaning $\mu_D(v) = 0$, has every far head protected.

A direction $a \rightarrow b$ on a missing pair is *convenient* if every original predecessor of a already reaches b in one or two steps. A missing pair is good if at least one direction is convenient, and bad otherwise. All such predicates carry an explicit graph index whenever the graph has changed. A *global median order* maximizes the total number of forward arcs among all vertex orders. Its last vertex is its feed. Local feedback inequalities alone do not justify the sedimentation step used later.

Remark 2.2 (Two kinds of unit). Gap one means $g_D(v) = 1$. Residual one means $\eta_D(v) = 1$. Actual failure-source status refers to a witnessed nonconvenient direction of an original bad pair. None of these three notions is identified with another without an explicit lemma.

3 Incoming-neighborhood pattern certificates

We first give a certificate that depends on a small induced subgraph but controls second neighborhoods in the original graph. Throughout this section, D is a finite nonempty oriented graph. Put

$$g_D(v) = |N_D^+(v)| - |N_D^{++}(v)|,$$

where $N_D^{++}(v)$ consists of the vertices at directed distance exactly two from v . A vertex with $g_D(v) \leq 0$ is called a Seymour vertex. No extremal or minimal-counterexample assumption is made in this section.

3.1 Patterns and the transfer theorem

Let H be a finite nonempty oriented graph. Define its family of *permitted incoming patterns* by

$$\mathcal{P}(H) = \{P \subseteq V(H) : P \subseteq N_H^-(j) \text{ for some } j \in V(H)\}. \quad (3.1)$$

In particular, the empty set belongs to $\mathcal{P}(H)$. For an arbitrary $P \subseteq V(H)$, write

$$R_H(P) = \{i \in V(H) \setminus P : N_H^+(i) \cap P \neq \emptyset\}.$$

Thus $R_H(P)$ consists of the vertices outside P that send an arc into P . If λ is a nonnegative vector indexed by $V(H)$, use the notation $\lambda(S) = \sum_{i \in S} \lambda_i$.

Definition 3.1 (Pattern certificate). A pattern certificate for H is a nonzero vector $\lambda \geq 0$ such that

$$\lambda(R_H(P)) \geq \lambda(P) \quad \text{for every } P \in \mathcal{P}(H). \quad (3.2)$$

The inequalities are homogeneous, so a certificate may always be normalized by $\lambda(V(H)) = 1$.

For a nonempty set $C \subseteq V(D)$, let $H = D[C]$ and associate to every original vertex v the pattern

$$\Pi_C(v) = N_D^-(v) \cap C.$$

We say that C is *incoming-covered* if

$$\Pi_C(v) \in \mathcal{P}(H) \quad (v \in V(D)). \quad (3.3)$$

The witness for this condition may depend on v ; the condition does not require any particular orientation between v and that witness.

For clarity, let $\mathbf{N}_D$ denote the matrix with rows and columns indexed by $V(D)$ and entries

$$(\mathbf{N}_D)_{uv} = \begin{cases} -1, & v \in N_D^+(u), \\ 1, & v \in N_D^{++}(u), \\ 0, & \text{otherwise.} \end{cases}$$

The row sum at u is $-g_D(u)$.

Theorem 3.2 (Transfer from an incoming-covered core). *Let $C \subseteq V(D)$ be nonempty and incoming-covered, and put $H = D[C]$. If λ is a pattern certificate for H , then its extension $\hat{\lambda}$ by zero to $V(D)$ satisfies*

$$\mathbf{N}_D^\top \hat{\lambda} \geq 0.$$

Consequently some $c \in C$ with $\lambda_c > 0$ is a Seymour vertex of D .

Proof. Fix an original target v and put $P = \Pi_C(v)$. The first predecessors of v in C are precisely the vertices of P . If $i \in R_H(P)$, choose $j \in N_H^+(i) \cap P$. Both arcs of the path $i \rightarrow j \rightarrow v$ are original arcs of D , and $i \notin P$ says that $i \rightarrow v$ is not an arc. Also $i \neq v$: otherwise $v \rightarrow j$ and $j \rightarrow v$ would form a digon. It follows that $v \in N_D^{++}(i)$.

The total λ -mass of the core vertices that have v as an exact second neighbor is therefore at least $\lambda(R_H(P))$. Since P is permitted, this is at least $\lambda(P)$, the total mass of its core first predecessors. This proves the asserted matrix inequality at the column indexed by v .

Summing over the columns gives

$$0 \leq \mathbf{1}^\top \mathbf{N}_D^\top \hat{\lambda} = - \sum_{c \in C} \lambda_c g_D(c).$$

Since λ is nonzero and nonnegative, not every vertex of its support can have positive gap. $\square$

The certificate weights the possible sources in C . The neighborhoods and their cardinalities in the conclusion remain those of the unweighted original graph. In particular, vertices outside C are retained as targets in every column of the transfer inequality.

3.2 Tournament cores

Tournament cores admit certificates of arbitrary order. We use the losing-density formulation underlying Fisher's proof for tournaments [7]. The short argument below also specifies the sign convention for the skew matrix.

Lemma 3.3 (A skew-game density). *Let B be a real skew-symmetric matrix. There is a nonnegative vector λ with $\mathbf{1}^\top \lambda = 1$ and $B\lambda \geq 0$.*

Proof. Let Δ be the probability simplex of the appropriate dimension. Finite minimax gives

$$\max_{\lambda \in \Delta} \min_{p \in \Delta} p^\top B\lambda = \min_{p \in \Delta} \max_{\lambda \in \Delta} p^\top B\lambda.$$

The left-hand side is at most zero, by choosing $p = \lambda$ in the inner minimum. The right-hand side is at least zero, by choosing $\lambda = p$ in the inner maximum. Both values are therefore zero. A maximizing λ satisfies $p^\top B\lambda \geq 0$ for every $p \in \Delta$, and in particular for every coordinate vector. $\square$

If A_H is the adjacency matrix of a tournament H , put $B_H = A_H - A_H^\top$. A vector supplied by Lemma 3.3 is called a losing density of H . At each vertex, its outgoing density is at least its incoming density.

Lemma 3.4 (Expansion of covered patterns in tournaments). *Every losing density of a nonempty tournament H is a pattern certificate for H .*

Proof. Fix $P \subseteq N_H^-(j)$ and set

$$R = R_H(P), \quad F = V(H) \setminus (P \cup R).$$

The witness j belongs to F , so F is nonempty. Tournament completeness implies that every arc between P and F is directed from P to F : a vertex of F has no arc into P by its definition.

First suppose that $\lambda(F) > 0$. Sum the nonnegative quantities $\lambda_i(B_H \lambda)_i$ over $i \in F$. The terms internal to F cancel by skew-symmetry. The contribution from pairs with the other endpoint in P is $-\lambda(F)\lambda(P)$. The contribution from pairs with the other endpoint in R is at most $\lambda(F)\lambda(R)$. Consequently

$$0 \leq \sum_{i \in F} \lambda_i(B_H \lambda)_i \leq \lambda(F)(\lambda(R) - \lambda(P)),$$

and division by $\lambda(F)$ proves the required inequality.

If $\lambda(F) = 0$, apply $(B_H \lambda)_j \geq 0$ at any $j \in F$. The incoming mass at j is at least $\lambda(P)$, whereas its outgoing mass is at most $\lambda(R)$. This again gives $\lambda(R) \geq \lambda(P)$. The argument also covers $P = \emptyset$. $\square$

Corollary 3.5 (Incoming-covered tournament cores). *If a nonempty incoming-covered set C induces a tournament in D , then C contains a Seymour vertex of D . More precisely, the zero extension of every losing density of $D[C]$ satisfies the matrix inequality in Theorem 3.2.*

Proof. Combine Lemmas 3.3 and 3.4 with Theorem 3.2. $\square$

3.3 Maximal incoming classes and exact small-core certificates

Define an equivalence relation on $V(D)$ by

$$u \sim_D v \iff N_D^-(u) = N_D^-(v).$$

The equivalence classes are partially ordered by containment of their common incoming sets. We call them *incoming classes*, and maximality will always refer to this inclusion order. Distinct members of one class cannot be adjacent, since an internal arc would put its tail in its own incoming set by equality of the two incoming neighborhoods. An incoming class need not have a common outgoing neighborhood.

Lemma 3.6 (Representative cores are incoming-covered). *Choose one representative from each maximal incoming class of D , and let C be the set of chosen vertices. Then C is incoming-covered.*

Proof. For every $v \in V(D)$, finiteness of the inclusion poset gives a maximal class with representative c such that $N_D^-(v) \subseteq N_D^-(c)$. Intersecting with C gives

$$\Pi_C(v) \subseteq N_D^-(c) \cap C = N_{D[C]}^-(c).$$

This is exactly the required condition. $\square$

Lemma 3.7 (Exact certificates for cores of order at most five). *Every nonempty oriented graph of order at most five has a pattern certificate. A certificate can be chosen with integer entries in $\{0, 1, 3\}$.*

Computer-assisted proof. The certificate data consist of one nonzero nonnegative integer vector for every labelled oriented graph of orders one through five. The encoding lists the unordered pairs

$$(0, 1), (0, 2), \dots, (0, n-1), (1, 2), \dots, (n-2, n-1)$$

in lexicographic order. The corresponding ternary digit is zero for a missing pair, one for the arc from the smaller endpoint to the larger endpoint, and two for the reverse arc. The first pair corresponds to the least significant digit. Thus the codes $0, \dots, 3^{\binom{n}{2}} - 1$ cover every labelled oriented graph exactly once.

For every code, the verifier reconstructs its adjacency matrix and checks that the supplied vector has length n , nonnegative integer entries, and positive sum. It then enumerates every subset $P \subseteq \{0, \dots, n-1\}$. A subset is retained precisely when some j satisfies $i \rightarrow j$ for every $i \in P$. For each retained subset, the verifier forms $R_H(P)$ directly from the reconstructed adjacency matrix and checks (3.2) with integer arithmetic.

The numbers of graph codes for $n = 1, 2, 3, 4, 5$ are, respectively,

$$1, \quad 3, \quad 27, \quad 729, \quad 59049.$$

All 59809 certificates pass all permitted-pattern inequalities. There are 522213 such checks, of which 462404 use a nonempty pattern. The largest integer entry is 3, and the largest sum of entries is 9. Among the order-five certificates, 59009 are uniform on their supports and the remaining 40 are nonuniform. This exhausts the finite assertion.

The complete tables and verification sources accompany the manuscript. In particular, `verify_core_pattern_certificates_independent.js` reconstructs Boolean adjacency matrices and checks the inequalities without calling an optimization solver or using a generator's graph utilities. The integer tables, rather than a floating-point optimization result, are the certificates used in this proof. $\square$

Theorem 3.8 (An entire Seymour maximal class). *Every finite nonempty oriented graph with at most five maximal incoming classes has a maximal incoming class all of whose vertices are Seymour vertices. There is no restriction on the total number of vertices or on the sizes of the classes.*

Proof. Suppose that no maximal incoming class consists entirely of Seymour vertices. Choose from each maximal class a vertex with positive gap, and let C be the set of chosen representatives. Then $1 \leq |C| \leq 5$, and Lemma 3.6 shows that C is incoming-covered. Lemma 3.7 supplies a pattern certificate for $D[C]$. Theorem 3.2 consequently gives a Seymour vertex in C , contrary to the selection of all its members. $\square$

Corollary 3.9. *Every counterexample to Seymour's second neighborhood conjecture has at least six maximal incoming classes.*

Proof. An all-deficient graph has no Seymour vertex, so Theorem 3.8 applies contrapositively. $\square$

The role of the computation is confined to the finitely many cores in Lemma 3.7. The passage to arbitrary orders and arbitrary class sizes in Theorem 3.8 uses the original-graph transfer theorem. It is not an enumeration of all graphs to which that theorem applies.

3.4 Strict duality and realization of an infeasible core

The pattern system also gives an equivalent formulation of the full conjecture. Its reverse implication requires a realization argument: infeasibility of a system on a core must yield an actual oriented graph with positive gap at every vertex.

Theorem 3.10 (Pattern feasibility and the second neighborhood conjecture). *The following statements are equivalent:*

1. *Every finite nonempty oriented graph has a Seymour vertex.*
2. *Every finite nonempty oriented graph H has a pattern certificate.*

More explicitly, if the normalized system

$$\lambda \geq 0, \quad \sum_i \lambda_i = 1, \quad \lambda(R_H(P)) \geq \lambda(P) \quad (P \in \mathcal{P}(H))$$

is infeasible for some H , a finite all-deficient oriented graph can be constructed from H and a rational strict dual solution.

Proof. First assume statement (2). For an arbitrary D , take the core to be D itself. Its pattern at a target v is the full set $N_D^-(v)$, which is permitted. Theorem 3.2 yields a Seymour vertex, proving statement (1).

For the converse, suppose that the normalized system for a nonempty core H is infeasible. For each permitted pattern, define the column vector

$$a_P(i) = \mathbf{1}_{i \in R_H(P)} - \mathbf{1}_{i \in P}.$$

On the probability simplex Δ indexed by $V(H)$, the continuous function $\min_{P \in \mathcal{P}(H)} \lambda^\top a_P$ is strictly negative everywhere. Compactness therefore gives

$$\max_{\lambda \in \Delta} \min_{P \in \mathcal{P}(H)} \lambda^\top a_P < 0. \quad (3.4)$$

By finite minimax, there are nonnegative real pattern weights whose weighted sum of the a_P has every coordinate strictly negative. They may be chosen rational: keep their zero coordinates equal to zero and approximate the positive coordinates sufficiently closely by positive rationals, using strictness of all the finitely many inequalities. Clearing denominators gives nonnegative integers w_P such that

$$\gamma_i = \sum_{P \in \mathcal{P}(H)} w_P (\mathbf{1}_{i \in P} - \mathbf{1}_{i \in R_H(P)}) > 0 \quad (i \in V(H)). \quad (3.5)$$

For each permitted pattern P , choose a witness $j(P)$ with $P \subseteq N_H^-(j(P))$. Fix a positive integer K . The new graph D_K has two kinds of vertices. For each $j \in V(H)$, it has a distinguished vertex c_j with label j and pattern $N_H^-(j)$. For each $P \in \mathcal{P}(H)$, it also has Kw_P vertices with pattern P and label $j(P)$. Distinct vertices are retained when patterns or labels coincide. If $\ell(v)$ and $P(v)$ denote the label and pattern of a vertex, define

$$v \rightarrow u \iff \ell(v) \in P(u). \quad (3.6)$$

Every pattern satisfies $P(v) \subseteq N_H^-(\ell(v))$, so $\ell(v) \notin P(v)$; hence (3.6) creates no loop. If both $v \rightarrow u$ and $u \rightarrow v$ held, the same containments would give both label arcs $\ell(v) \rightarrow \ell(u)$ and $\ell(u) \rightarrow \ell(v)$ in H . For unequal labels this is a digon, and for equal labels it is a loop. Thus D_K is oriented. Its distinguished vertices induce exactly H .

Consider any source v with label i . Its first targets are exactly the vertices u with $i \in P(u)$. We claim that its exact second targets are exactly those with $i \in R_H(P(u))$. Indeed, if $v \rightarrow x \rightarrow u$ and $i \notin P(u)$, put $j = \ell(x)$. The first arc and $P(x) \subseteq N_H^-(j)$ imply $i \rightarrow j$ in H , while the second arc says $j \in P(u)$. Thus $i \in R_H(P(u))$. Conversely, if $i \in R_H(P(u))$, choose $j \in N_H^+(i) \cap P(u)$. The distinguished vertex c_j supplies the path $v \rightarrow c_j \rightarrow u$, and $i \notin P(u)$ excludes a first arc. The target cannot be v : since $P(v) \subseteq N_H^-(i)$, such a j would give a digon $i \leftrightarrow j$ in H .

The extra pattern vertices therefore contribute $K\gamma_i$ to the gap of every source labelled i . The distinguished vertices contribute $g_H(i)$, because they supply exactly the first and exact second targets of i in H . We have proved the identity

$$g_{D_K}(v) = K\gamma_i + g_H(i) \quad \text{whenever } \ell(v) = i. \quad (3.7)$$

Choose K so large that the right-hand side is positive for every $i \in V(H)$. This is possible by (3.5). Then D_K is a finite all-deficient oriented graph. Consequently statement (1) rules out infeasibility for every H , proving statement (2). $\square$

Corollary 3.11 (An equivalent whole-class formulation). *Seymour's second neighborhood conjecture is equivalent to the assertion that every finite nonempty oriented graph has a maximal incoming class consisting entirely of Seymour vertices.*

Proof. The whole-class assertion plainly implies the conjecture. Conversely, assume the conjecture, and hence universal pattern feasibility by Theorem 3.10. If each maximal incoming class of D contained a positive-gap vertex, choose one such representative from each class. Lemma 3.6 and Theorem 3.2 would give a Seymour vertex among these representatives, a contradiction. $\square$

3.5 Lifting through iterated representative cores

The representative operation can be repeated, and it preserves more than the existence of one Seymour vertex: a certificate at a later stage lifts to a certificate at every preceding stage.

Lemma 3.12 (Lifting a pattern certificate). *Let C contain one representative from each maximal incoming class of an oriented graph H , and put $K = H[C]$. The extension by zero of any pattern certificate for K is a pattern certificate for H .*

Proof. Fix $P \subseteq N_H^-(v)$. Choose a representative $c \in C$ such that $N_H^-(v) \subseteq N_H^-(c)$. Then

$$P \cap C \subseteq N_K^-(c), \quad R_K(P \cap C) \subseteq R_H(P) \cap C.$$

For the second inclusion, a vertex of C outside $P \cap C$ is outside P , and an arc into $P \cap C$ is also an arc of H into P . If λ is the certificate on C and $\hat{\lambda}$ is its zero extension, it follows that

$$\hat{\lambda}(R_H(P)) \geq \lambda(R_K(P \cap C)) \geq \lambda(P \cap C) = \hat{\lambda}(P).$$

The lifted vector is nonnegative and nonzero, so it is a certificate. $\square$

Corollary 3.13 (Iterated-core sufficient conditions). *Starting from D , repeatedly take the induced subgraph on one representative from each maximal incoming class of the current graph. If any resulting core has at most five vertices, or is a tournament of any order, then the original graph D has a Seymour vertex in that core.*

Proof. The small-core certificate lemma or the tournament certificate lemma supplies a certificate at the specified stage. Apply Lemma 3.12 successively in reverse order. This gives a certificate on D supported in the specified core. Applying Theorem 3.2 with the whole graph as its core gives an original Seymour vertex in that support. $\square$

Corollary 3.14 (Reduction to incomparable incoming neighborhoods). *Universal pattern feasibility, and therefore Seymour's second neighborhood conjecture, is equivalent to pattern feasibility for oriented graphs whose incoming neighborhoods are distinct and pairwise incomparable under inclusion.*

Proof. Whenever the representative operation reduces the vertex set, repeat it. The process terminates because the order decreases. If the operation no longer reduces the order, every incoming class is a singleton and every class is maximal. The incoming neighborhoods are therefore distinct and pairwise incomparable. A certificate for this terminal graph lifts to the original graph by Lemma 3.12. The reverse implication is immediate. $\square$

This is an induced-subgraph reduction: it deletes vertices and does not add arcs. Its conclusion concerns the original neighborhoods of vertices in the final core. For a single chosen chain it does not assert that an entire original maximal incoming class consists of Seymour vertices; the whole-class conclusions above use representative selection under the negation of that stronger assertion.

4 Predecessor containment and extremal counterexamples

In this section we study a surgery that preserves positive gaps by copying outgoing neighborhoods under incoming-neighborhood containment. The surgery itself is valid in an arbitrary finite oriented graph. Its structural consequences will explicitly state whether they require minimum order or a stronger lexicographic extremal choice.

We call D *all-deficient* if $g_D(v) > 0$ for every vertex. Such a graph is a counterexample to the second neighborhood conjecture. The gap-one predecessor and gap-one cycle phenomena occur in the minimal-counterexample analysis of Brantner et al. [2]. Minimum-order reductions also play a central role in [5]. The arguments below give statements about entire incoming classes and establish the precise extremal hypotheses needed for an independent-module normal form.

4.1 Copying one row and copying all rows

All containment predicates and all copied neighborhoods in the next two results are evaluated in the original graph D .

Lemma 4.1 (One-row copying under predecessor containment). *Let $a \neq b$ and suppose that $N_D^-(a) \subseteq N_D^-(b)$. Replace the outgoing neighborhood of a by $N_D^+(b)$, leaving every other outgoing neighborhood unchanged, and denote the resulting graph by $D^{a \leftarrow b}$. Then this graph is oriented and*

$$N_{D^{a \leftarrow b}}^+(a) = N_D^+(b), \quad N_{D^{a \leftarrow b}}^{++}(a) = N_D^{++}(b).$$

For every $v \neq a$,

$$N_{D^{a \leftarrow b}}^+(v) = N_D^+(v), \quad N_{D^{a \leftarrow b}}^{++}(v) \subseteq N_D^{++}(v).$$

In particular, the new gap at a is $g_D(b)$, all other gaps do not decrease, and an all-deficient graph remains all-deficient.

Proof. There is no original arc $b \rightarrow a$: it would put b in $N_D^-(a)$ and hence in $N_D^-(b)$, contrary to the absence of loops. Nor is there an original two-step path $b \rightarrow x \rightarrow a$, since $x \rightarrow a$ would imply $x \rightarrow b$ and form a digon with $b \rightarrow x$. Thus a is neither a first nor a second target of b .

Every $x \in N_D^+(b)$ is distinct from a and cannot point to a , again because $x \rightarrow a$ would imply $x \rightarrow b$. The new row at a therefore creates neither a loop nor a digon. The intermediate vertices in $N_D^+(b)$ retain their outgoing neighborhoods. Their union of outgoing targets contains neither a , by the two-step exclusion just proved, nor b , by the absence of digons. Subtracting the copied first set $N_D^+(b)$ consequently gives exactly $N_D^{++}(b)$ as the new second neighborhood of a .

For $v \neq a$, only a two-step path using the changed row could produce a new second target. Such a path has the form $v \rightarrow a \rightarrow x$, where $v \rightarrow a$ and $b \rightarrow x$ are original arcs. Incoming containment gives the original arc $v \rightarrow b$, so $v \rightarrow b \rightarrow x$ was already an original two-step path. The first set at v is unchanged, so no new exact second target appears. Some second targets may disappear. The gap statements now follow by counting the first and second sets. $\square$

Theorem 4.2 (A simultaneous predecessor map). *Let $\rho : V(D) \rightarrow V(D)$ satisfy*

$$N_D^-(v) \subseteq N_D^-(\rho(v)) \quad (v \in V(D)). \quad (4.1)$$

Define D^ρ by simultaneously setting

$$N_{D^\rho}^+(v) = N_D^+(\rho(v)) \quad (v \in V(D)).$$

Then D^ρ is oriented, and for every v ,

$$N_{D^\rho}^{++}(v) \subseteq N_D^{++}(\rho(v)), \quad g_{D^\rho}(v) \geq g_D(\rho(v)). \quad (4.2)$$

The map ρ need not be injective, and its image may contain vertices whose rows are also changed. In particular, all-deficiency is preserved.

Proof. A new loop at v would mean an original arc $\rho(v) \rightarrow v$. By (4.1), this would imply an original loop at $\rho(v)$. If $v \rightarrow u$ and $u \rightarrow v$ were both new arcs, the original arcs $\rho(v) \rightarrow u$ and $\rho(u) \rightarrow v$, together with containment at u and v , would give the original label arcs $\rho(v) \rightarrow \rho(u)$ and $\rho(u) \rightarrow \rho(v)$. These form a digon when the two images differ and a loop when they coincide. Thus the new graph is oriented.

Consider a new two-step path $v \rightarrow u \rightarrow x$. It means that $\rho(v) \rightarrow u$ and $\rho(u) \rightarrow x$ are original arcs. Applying incoming containment at u to the first of these arcs yields the original arc $\rho(v) \rightarrow \rho(u)$. Hence

$$\rho(v) \rightarrow \rho(u) \rightarrow x$$

is an original two-step path. Its intermediate vertex cannot equal $\rho(v)$, since that would require a loop, and its endpoint cannot equal $\rho(v)$, since that would require a digon. If x is an exact second target of v in D^ρ , then $x \notin N_{D^\rho}^+(v) = N_D^+(\rho(v))$. Therefore it is an original exact second target of $\rho(v)$, proving the inclusion in (4.2). The first neighborhood has exactly the copied original size, so the gap inequality follows. $\square$

Corollary 4.3 (Copying a prescribed subset). *Let $A \subseteq V(D)$, and choose $\rho(a)$ for each $a \in A$ with $N_D^-(a) \subseteq N_D^-(\rho(a))$. Extend ρ by the identity outside A . Then simultaneous copying gives*

$$g_{D^\rho}(a) \geq g_D(\rho(a)) \quad (a \in A), \quad g_{D^\rho}(v) \geq g_D(v) \quad (v \notin A).$$

Proof. The extended map satisfies (4.1) at every vertex, so Theorem 4.2 applies. $\square$

The simultaneous formulation is essential for coupled choices of the map: all containments refer to one graph, and all rows are copied from that same graph. A sequence that recomputes containment after each individual copy is a different procedure and is not needed here.

4.2 A unit-predecessor cover at minimum order

For the remainder of this subsection, assume that D is all-deficient and has the smallest possible number of vertices among all all-deficient oriented graphs. This is a global minimum-order assumption. For $f \in V(D)$ define

$$A_f = \{a \in V(D) : a \rightarrow f \text{ and } g_D(a) = 1\}.$$

The next obstruction concerns a simultaneous choice of replacement vertices outside the set being copied.

Lemma 4.4 (Unit predecessors cannot all be copied outside their set). *For no vertex f is there a map*

$$\rho : A_f \longrightarrow V(D) \setminus A_f \quad \text{such that} \quad N_D^-(a) \subseteq N_D^-(\rho(a)) \quad (a \in A_f).$$

In particular $A_f \neq \emptyset$ for every f .

Proof. Suppose such a map exists. Extend it by the identity outside A_f and copy the rows simultaneously. The resulting graph D^ρ is all-deficient by Theorem 4.2.

We claim that every predecessor of f in D^ρ has gap at least two. An unchanged predecessor was originally a predecessor outside A_f , and hence had positive integral gap at least two; its gap has not decreased. If a changed vertex $a \in A_f$ still points to f , then its chosen image $b = \rho(a)$ originally pointed to f . Since $b \notin A_f$, we again have $g_D(b) \geq 2$, and the copy inequality gives $g_{D^\rho}(a) \geq 2$. There are no other new predecessors because all rows outside A_f were unchanged.

Delete f from D^ρ . Each of its predecessors loses one first target and can only lose exact second targets among the remaining vertices; its gap therefore remains at least one. Every other remaining vertex loses no first target and acquires no exact second target, so its gap also remains positive. The deleted graph is nonempty: an all-deficient oriented graph has positive outdegree at every vertex and therefore at least three vertices. We have produced a smaller all-deficient graph, contradicting minimum order.

If A_f were empty, the empty map would meet the stated conditions and the same deletion argument would apply. $\square$

Theorem 4.5 (Every vertex receives a maximal pure-unit class). *For every $f \in V(D)$, some maximal incoming class C satisfies*

$$C \subseteq A_f.$$

Equivalently, every member of C has original gap one and sends an original arc to f .

Proof. Suppose no maximal incoming class is contained in A_f . Each maximal class then has a representative outside A_f . For every $a \in A_f$, choose a maximal class whose incoming set contains $N_D^-(a)$, and let $\rho(a)$ be its chosen representative outside A_f . Such a maximal class exists in the finite inclusion poset. The resulting map satisfies

$$N_D^-(a) \subseteq N_D^-(\rho(a)), \quad \rho(a) \notin A_f \quad (a \in A_f),$$

contradicting Lemma 4.4. $\square$

Call a maximal incoming class *pure-unit* if all its members have gap one, and let $\mathcal{C}_1$ be the family of these classes.

Corollary 4.6 (A complete-arc cycle of pure-unit classes). *There are at least three classes in $\mathcal{C}_1$. Define an oriented quotient on $\mathcal{C}_1$ by placing an arc $C \rightarrow C'$ when every vertex of C points to every vertex of C' . This quotient has positive minimum indegree and contains a directed cycle of length at least three. Selecting one vertex from each class on such a cycle gives an original directed cycle all of whose vertices have gap one.*

Proof. Theorem 4.5 makes $\mathcal{C}_1$ nonempty. Fix $C' \in \mathcal{C}_1$ and $f \in C'$. The theorem gives $C \in \mathcal{C}_1$ with every member of C pointing to f . Every vertex of C' has the same incoming neighborhood as f ; hence all these arcs extend to complete arcs from C to C' . The classes are distinct, since an incoming class is independent. Thus each quotient vertex has an incoming arc.

The quotient has no loops or digons, by its definition and the orientedness of D . Following predecessors in this finite quotient produces a directed cycle. Such a cycle has at least three vertices. Its complete interclass arcs allow any choice of one original vertex from each cycle class, proving the last assertion. $\square$

Combining this with Corollary 3.9, a minimum-order counterexample would have at least six maximal incoming classes, at least three of which are pure-unit. Neither assertion says that every maximal incoming class must be pure-unit.

4.3 A lexicographic extremal normal form

Suppose a counterexample exists. Among all all-deficient oriented graphs, choose D according to the following successive objectives:

1. minimize $|V(D)|$;
2. subject to the first objective, minimize $|A(D)|$;
3. subject to the first two objectives, maximize $\sum_{v \in V(D)} g_D(v)$;
4. subject to the first three objectives, maximize

$$\Psi(D) = \sum_F |\{v \in V(D) : N_D^+(v) = F\}|^2,$$

where the sum ranges over the distinct outgoing neighborhoods.

These choices exist conditionally on the existence of a counterexample: the first minimum is a positive integer, and after fixing it only finitely many labelled oriented graphs remain. The second objective ranges over all counterexamples of that order, rather than just subgraphs of one chosen graph. Call a graph selected in this way *globally extremal*.

Lemma 4.7. *A globally extremal counterexample is strongly connected and minimal under deletion of any nonempty set of arcs while the vertex set is fixed.*

Proof. If it were not strongly connected, choose a sink strongly connected component. It is a proper nonempty vertex subset with no outgoing arc to its complement. Both first and second neighborhoods of its vertices therefore remain unchanged in the induced subgraph. That subgraph would be all-deficient, contrary to minimum order. If deleting a nonempty arc set preserved all-deficiency, the resulting graph would have the same order and fewer arcs, contrary to the second objective. $\square$

Proposition 4.8 (Degree and gap monotonicity under incoming containment). *In a globally extremal counterexample, if $a \neq b$ and $N_D^-(a) \subseteq N_D^-(b)$, then*

$$d_D^+(a) \leq d_D^+(b).$$

If these outdegrees are equal, then $g_D(a) \geq g_D(b)$. In particular, equal incoming neighborhoods imply equal outdegrees and equal gaps. If in addition $a \rightarrow b$, the missing degrees satisfy

$$\mu_D(a) \geq \mu_D(b) + 1, \quad \mu_D(v) = |V(D)| - 1 - d_D^+(v) - d_D^-(v).$$

Proof. By Lemma 4.1, copying the outgoing row of b to a preserves all-deficiency. It changes the arc count by $d_D^+(b) - d_D^+(a)$. A negative change would contradict the second extremal objective, which proves the degree inequality.

If the degrees coincide, the copied graph has the same order and arc count. Its new gap at a is $g_D(b)$, and all other gaps do not decrease. Its total gap is thus at least the old total plus $g_D(b) - g_D(a)$. The third objective excludes a positive increase, giving $g_D(a) \geq g_D(b)$. When the incoming neighborhoods are equal, apply both conclusions in both directions.

If $a \rightarrow b$, then $a \in N_D^-(b) \setminus N_D^-(a)$, so incoming containment yields $d_D^-(b) \geq d_D^-(a) + 1$. Therefore

$$\mu_D(a) - \mu_D(b) = d_D^+(b) - d_D^+(a) + d_D^-(b) - d_D^-(a) \geq 1.$$

□

Theorem 4.9 (Independent incoming classes in an extremal graph). *In a globally extremal counterexample,*

$$N_D^-(a) = N_D^-(b) \implies N_D^+(a) = N_D^+(b).$$

Consequently every incoming class is an independent module: each outside vertex has uniform adjacency and orientation to the entire class. Members of a class have the same first and exact second neighborhoods, the same gap, and the same missing degree.

Proof. Suppose that a and b have the same incoming neighborhood but different outgoing neighborhoods F_a and F_b . Proposition 4.8 gives equal outdegrees and equal gaps. Both single-row copying directions are permitted. They preserve the first two extremal objectives; their total gaps cannot decrease, and by the third objective cannot increase. Thus both operations also preserve the third objective.

Let r_a and r_b be the numbers of vertices whose outgoing neighborhoods are F_a and F_b , respectively. Interchange the names if necessary so that $r_a \leq r_b$, and copy b 's row to a . Every other outgoing row is unchanged. The value of the fourth objective changes by

$$(r_a - 1)^2 + (r_b + 1)^2 - r_a^2 - r_b^2 = 2(r_b - r_a) + 2 > 0.$$

This remains the correct formula if the old row class disappears. The increase contradicts the fourth extremal objective, proving equality of the outgoing neighborhoods.

An incoming class is independent, as noted above. Equality of both incoming and outgoing neighborhoods gives uniform adjacency to every outside vertex, including uniform nonadjacency when no arc is present. The unions of outgoing neighborhoods over the common first set are identical for all members of the class. Such a union contains no member of the class: a path $a \rightarrow x \rightarrow b$ between two class members would imply $x \rightarrow a$ from their equal incoming sets and hence a digon. Subtracting the common first set therefore gives identical exact second sets. Equality of the gaps and missing degrees follows. □

The independent-module conclusion concerns the selected globally extremal graph. The row-copying inequalities hold without that selection, but the degree ordering, gap ordering, and final symmetrization use its successive objectives. In particular, arc-deletion minimality alone does not supply the tie-breaking properties used in Proposition 4.8 and Theorem 4.9.

5 Protection closure and median feeds

The incoming-core reduction takes induced subgraphs. We now keep the vertex set fixed and add arcs. Every convenience predicate in this section refers to the graph present at the stated stage.

5.1 Two safe operations

A missing direction $a \rightarrow b$ is *forced* in H if some vertex w satisfies $a \triangleleft_H w$ and $w \rightarrow b \in A(H)$.

Lemma 5.1. *Forced directions are consistent, uniquely convenient in H , and mandatory in every tournament completion preserving all protections of H . Adding all forced directions simultaneously preserves every old protection.*

Proof. For a witness $a \triangleleft_H w$, $w \rightarrow b$, every predecessor p of a also points to w , and hence reaches b within two steps. Thus $a \rightarrow b$ is convenient. Its reverse fails at w , since w cannot reach its protected far head a . Opposite directions cannot both be forced: each would have been proved convenient while the other witness proves it nonconvenient. In a tournament, $a \triangleleft w$ also gives outgoing domination, so $w \rightarrow b$ forces $a \rightarrow b$.

Fix an old $y \triangleleft_H x$. An old predecessor of y already points to x . For a newly incoming $a \rightarrow y$, choose its witness $a \triangleleft_H w$, $w \rightarrow y$. Then $w \rightarrow x$ originally. If ax is missing it is forced $a \rightarrow x$ in the same round. If $x \rightarrow a$ were an old arc, protection at a, w would give $x \rightarrow w$, a digon with $w \rightarrow x$. Thus the required new incoming nesting is preserved in every case. $\square$

There is a second, sequential operation. If ab is missing, $N_H^-(a) \subseteq N_H^-(b)$, and b has no protection successor in H , add the single arc $a \rightarrow b$. Call this a *weak step*.

Lemma 5.2. *A weak step is convenient in its preceding graph, preserves all old protections, and creates $a \triangleleft b$.*

Proof. Every old predecessor of a already points to b . Only $N^-(b)$ changes. An old relation $y \triangleleft b$ becomes easier to satisfy, and an old relation $b \triangleleft x$ is excluded by the hypothesis on b . All other incoming comparisons are unchanged. The added arc together with the old containment gives $a \triangleleft b$. $\square$

Starting at D , repeatedly perform a full forced round; when no forced direction remains, perform one available weak step and recompute. Stop when neither operation is available, obtaining a partial graph Q . Any deterministic choice among eligible weak steps may be used. The process terminates after at most $M(D)$ new pairs. Protections acquired at any stage persist. In particular, every added arc has a tail with a protection successor in Q . Simultaneous arbitrary weak choices are not part of the construction.

5.2 A fixed partial graph and a complete reference family

Since Q has no remaining forced pair, each $y \triangleleft_Q x$ has outgoing as well as incoming nesting. Indeed, $x \rightarrow z$ excludes $z \rightarrow y$ by incoming nesting; if yz were missing, $y \rightarrow z$ would

still be forced. A two-position swap therefore shows that every global median order of Q extends protection: reversing x, y gains their mutual arc and loses no contribution from an intermediate vertex. Missing arcs contribute zero.

Complete every remaining missing pair forward in a linear extension of the protection order of Q . Every such tournament preserves protection. To see this, consider $y \triangleleft_Q x$ and a newly incoming $a \rightarrow y$. The reference has $a < y < x$. An old $x \rightarrow a$ would have forced $y \rightarrow a$, contrary to terminality and the missing pair ay . Thus $a \rightarrow x$ is either old or is also completed forward. This proves the incoming nesting.

Let T_* maximize the ordinary median score over the family of *all* these extensions. A remaining pair of Q is called free.

Theorem 5.3 (Rigidity and original neighbourhoods). *Every global median of T_* respects protection in Q and puts every free arc forward. At any feed f of such an order,*

$$N_{T_*}^+(f) = N_Q^+(f) = N_D^+(f), \quad N_Q^{++}(f) = N_D^{++}(f).$$

If r is the number of free pairs, the maximum tournament median score is $\text{med}(Q) + r$, and the possible feeds across the maximizing completions are exactly the median feeds of Q .

Proof. Tournament domination makes every median respect protection. If a free arc were backward in one median of T_* , use that median as a new reference. This fixes the arcs of Q and turns all backward free arcs forward, strictly increasing the score. This contradicts maximality and applies to every median of the same maximizing tournament. The argument requires the entire extension family.

A feed has no outgoing free arc, since it would be backward. Nor can it be the tail of an arc added during closure: its persistent protection successor would have to occur later. Thus its first set was unchanged at every stage. Induct over the stages. Any new two-step path at f uses an original first arc and a newly added second arc. Convenience in the preceding graph says that this original predecessor f already reached the head within two steps. A first neighbour stays first throughout, so no new exact second head occurs. Old exact second paths persist because the first set is fixed.

A median of Q is an admissible reference and scores $\text{med}(Q) + r$ after completion. No order in any reference tournament can exceed this number: its Q contribution is at most $\text{med}(Q)$ and its free contribution at most r . Equality forces both bounds to be attained, proving the score and feed assertions. $\square$

5.3 Equality sedimentation and the original far budget

For a median order L of a tournament with feed f , a vertex $y \in N^-(f)$ is good if some $a \in N^+(f)$ occurs before y in L and has $a \rightarrow y$. Let G_L be the good set. The strong feed theorem of Havet and Thomassé gives

$$|N^+(f)| \leq |G_L| \leq |N^{++}(f)|.$$

When $|G_L| = |N^+(f)|$, their global sedimentation lemma replaces L by the order consisting of its bad vertices, then f , then $N^+(f) \cup G_L$, preserving each block's previous relative order. It remains a global median of the same tournament [12]. Each vertex of the last block moves strictly later. These statements concern ordinary global medians.

Theorem 5.4 (Strict original surplus). *Let f be a median feed of T_* with $g_D(f) > 0$, and set*

$$A_f = N_{T_*}^{++}(f) \setminus N_D^{++}(f), \quad R_f = \mathcal{M}_Q(f), \quad m_f = |R_f|.$$

Then $m_f > 0$ and $t_D(f) \geq g_D(f) + 1$. If $g_{T_*}(f) = 0$, then

$$\emptyset \neq R_f \subseteq U_{T_*}(f) \cap (\mathcal{F}_D(f) \setminus A_f), \quad t_D(f) \geq g_D(f) + m_f.$$

No all-deficiency or minimality assumption on the other vertices is needed.

Proof. Original protection survives. By Theorem 5.3,

$$N_{T_*}^{++}(f) = N_D^{++}(f) \dot{\cup} A_f, \quad A_f \subseteq \mathcal{F}_D(f), \quad g_{T_*}(f) = g_D(f) - |A_f| \leq 0.$$

If R_f were empty, f would be whole in Q . Every far head of a whole vertex is protected in that graph. Preservation of these protections would prevent any further second head at f , giving $g_{T_*}(f) = g_D(f) > 0$. This contradiction proves $m_f > 0$.

If the completed gap is zero, strong feed equality gives $G_L = N_{T_*}^{++}(f)$. A partner $r \in R_f \cap G_L$ has a free incoming arc $r \rightarrow f$, since the first set at f is original. Sedimentation puts f before r while keeping a median, contradicting free-arc rigidity. Thus every remaining partner is far in T_* . It was originally missing, so it cannot be an originally protected head, and old exact second paths survive. Hence it lies in $\mathcal{F}_D(f) \setminus A_f$. The two disjoint sets have sizes m_f and $g_D(f)$. If the completed gap is negative instead, integrality gives $|A_f| \geq g_D(f) + 1$ directly. $\square$

Theorem 5.5 (An original partner cycle). *Under the zero-gap hypotheses of Theorem 5.4, every vertex of $Q[R_f]$ has an incoming neighbour there, and the original graph $D[R_f]$ contains a directed cycle. Consequently*

$$m_f \geq 3, \quad t_D(f) \geq g_D(f) + m_f \geq g_D(f) + 3.$$

Proof. The feed is maximal in terminal protection. For $r \in R_f$, terminality of the weak operation gives $p \in N_Q^-(r) \setminus N_Q^-(f)$. Since r is far from f in T_* , $p \rightarrow r$ implies $p \rightarrow f$ there. Neither direction on pf was present in Q , so $p \in R_f$. Thus each remaining partner has a predecessor in the induced terminal graph.

Fix the final set R_f throughout the construction. For a forced internal addition $a \rightarrow b$, let $a \triangleleft_H w$, $w \rightarrow b$ be its preceding witness. Far-ness of b in T_* gives $w \rightarrow f$ there. If wf were adjacent in Q , its direction would already be $w \rightarrow f$, and the retained protection $a \triangleleft w$ would force the still-missing $a \rightarrow f$. Therefore $w \in R_f$. The internal addition shortcuts an old internal two-step path. All simultaneous such shortcuts preserve acyclicity of an acyclic induced graph.

For a single weak internal addition $a \rightarrow b$, creating the first cycle would require an old internal path $b \rightsquigarrow p \rightarrow a$. It cannot have length one, because ab was missing. Old incoming containment gives $p \rightarrow b$, already completing an old cycle. Thus weak steps cannot create the first cycle either. An acyclic $D[R_f]$ would remain acyclic, whereas a finite graph of positive minimum indegree has a cycle. Orientedness excludes cycle lengths one and two. $\square$

The stronger +3 bound applies to zero completed gap only. Omitting all weak steps still gives Theorem 5.4, but does not give the partner-cycle conclusion. The original cycle in the theorem is not a claim that every terminal cycle lifts edge by edge, nor that its vertices are actual bad-direction sources.

Corollary 5.6. *Every oriented graph with $t_D(v) \leq 1$ at every vertex has a Seymour vertex. More generally this holds if $t_D(v) \leq g_D(v)$ at every deficient vertex.*

Proof. Otherwise every original gap is positive and any feed contradicts the strict-surplus theorem. $\square$

If $h_D(f)$ counts the protected original far heads, partitioning the other vertices gives $t_D(f) - g_D(f) = n - 1 - 2d_D^+(f) - h_D(f)$. Hence every original-positive-gap closure feed satisfies

$$2d_D^+(f) + h_D(f) \leq n - 2. \quad (5.1)$$

At $n = 2\delta^+(D) + 3$, such a feed has minimum original outdegree and at most one protected far head. This locates a possibly deficient feed; it does not assert that a minimum-degree vertex already has the SNP in D .

5.4 The negative-gap equality branch

Every new original-far head belongs to G_L : its second arc cannot lie in Q , and therefore is free and forward, putting its original first intermediate before the head. Write $b_L = |N_D^{++}(f) \setminus G_L|$ and $s_L = |G_L| - d_D^+(f)$. Then

$$s_L = |A_f| - g_D(f) - b_L \geq 0. \quad (5.2)$$

When $s_L = 0$, sedimentation excludes $R_f \cap G_L$. Its second partners number at most $|A_f| - g_D(f)$ and are original nongood second vertices; its far partners number at most $t_D(f) - |A_f|$. Thus

$$|R_f| \leq t_D(f) - g_D(f) \quad (s_L = 0). \quad (5.3)$$

This includes negative completed gap. Put $R_2 = R_f \cap N_D^{++}(f)$, $R_\infty = R_f \cap U_{T^*}(f)$, and $U = R_\infty \setminus \text{Reach}_{D[R_f]}(R_2)$, where reachability includes the seed vertices. The supplement proves that closure creates no new reachability from R_2 into R_∞ inside R_f . If the whole set U is nonempty, its original induced graph $D[U]$ contains a directed cycle (Theorem S44.1). When $s_L > 0$, the cited equality-sedimentation move is unavailable. This does not assert that every median order of the tournament has the same slack.

6 Residual consequences in minimal counterexamples

Throughout this section, unless stated otherwise, D is in the MIN setting of Section 2. Put

$$\eta(v) = \mu(v) - 1 + d^{--}(v) - d^{++}(v), \quad R = \sum_v \eta(v) = 2M - n.$$

The last identity counts exact directed distance-two pairs by their two endpoints. Unlike the core certificate, the inequalities below use original arc-set minimality.

6.1 The original boundary and partner estimates

We collect the local estimates used here. Full proofs are given in the supplement so that the external-boundary convention is explicit.

Lemma 6.1 (Original deletion estimates). *Every original gap belongs to $\{1, 2\}$. If $\partial S \neq \emptyset$, then $|\partial^2 S| < |\partial S|$. Moreover $\eta(v) \geq 0$ for every v , and*

$$\eta(v) = 2\mu(v) + g(v) - 1 - |P(v)|.$$

Fix v with $P(v) \neq \emptyset$, and write

$$\begin{aligned} C &= \partial P(v), & X &= \partial^2 P(v), & J &= N^{--}(v) \setminus C, \\ B &= J \cap \mathcal{M}(v), & k &= |B|, & s &= |J \cap N^+(v)|. \end{aligned}$$

Delete the s arcs from v into J , obtaining $\widehat{D}$. Then

$$N_{\widehat{D}}^{++}(v) \subseteq E = (\mathcal{M}(v) \setminus B) \dot{\cup} X, \quad L = E \setminus N_{\widehat{D}}^{++}(v)$$

satisfies

$$|L| \leq \eta(v) - 2k - g(v)\mathbf{1}_{\{s>0\}}, \quad \bigcup_{r \in \mathcal{M}(v) \setminus B} (N^+(r) \cap U(v)) \subseteq L. \quad (6.1)$$

Every exceptional partner $r \in B$ admits the convenient direction $r \rightarrow v$. Residual zero gives $t(v) = 0$, residual one gives $t(v) \leq 1$, and $\eta(v) = t(v) = 1$ gives $P(v) \neq \emptyset$ and $N^{--}(v) = \partial P(v)$.

Proof. The gap bound follows by deleting one outgoing arc: other first sets are unchanged and their second sets can only shrink, while the deleted tail's gap decreases by at most two. The external-boundary deletion argument is Lemma S3.1; the original target partition yields Theorem S2.1 and the residual identity. The shared remainder estimate is Lemma S8.9; it subtracts the original second degree if $s = 0$, and the second degree forced by minimality after the nonempty deletion if $s > 0$. Its union bound counts distinct heads. Zero-residual protection and the unit bound are proved in the supplementary local-budget section. At residual one, a nonempty deletion gives $t = 0$; otherwise equality forces J empty. The empty- P identity gives a whole vertex and $t = 0$, proving the final assertion when $t = 1$. $\square$

The set L concerns an original deletion. It is never recomputed in a closure graph and then assigned the original minimality hypothesis. In particular, an exceptional missing far head can lie in X ; no inclusion $X \subseteq N^-(v)$ is assumed.

6.2 A residual-four threshold for zero completed gap

Let $S_v = \mathcal{M}(v) \cap U(v)$. For nonempty $P(v)$,

$$|S_v| \leq k + |L| \leq \eta(v) - k - g(v)\mathbf{1}_{\{s>0\}}. \quad (6.2)$$

Indeed, a nonexceptional missing far head belongs to E and cannot become reachable when arcs are deleted, so it lies in L . When $k = 0$, in fact $\mathcal{F}(v) \subseteq L$. A far incoming head y that is not protected has a predecessor r not pointing to v . It cannot be an outneighbour of v , since that would reach y in two steps. Thus it is a missing partner, and the union bound in (6.1) applies.

Theorem 6.2 (Zero-gap residual threshold). *At a zero-gap feed f of the weak-closure completion, let $m = |\mathcal{M}_Q(f)|$. Then*

$$\eta_D(f) \geq m + 1 \geq 4.$$

If $\eta_D(f) = 4$, its original packet necessarily satisfies

$$P_D(f) \neq \emptyset, \quad m = 3, \quad k = s = 0, \quad g_D(f) = 1, \quad t_D(f) = 4.$$

Furthermore $|P_D(f)| = 2\mu_D(f) - 4 \geq 2$, $N_D^{--}(f) = \partial P_D(f)$, $L = \mathcal{F}_D(f)$, and $|\partial^2 P_D(f)| = |\partial P_D(f)| - 1$.

Proof. Theorem 5.5 gives $m \geq 3$, $\mathcal{M}_Q(f) \subseteq S_f$, and $t_D(f) \geq g_D(f) + m$. If $P(f)$ is empty, the residual identity gives $\eta(f) = 2\mu(f) + g(f) - 1 \geq 2m + g(f) - 1 \geq m + 1$. For nonempty $P(f)$ and $k > 0$, (6.2) gives $\eta(f) \geq m + k + g(f)\mathbf{1}_{\{s>0\}} \geq m + 1$. For $k = 0$, the inclusion $\mathcal{F}(f) \subseteq L$ gives $\eta(f) \geq g(f) + m + g(f)\mathbf{1}_{\{s>0\}} \geq m + 1$.

At residual four these bounds force $P(f) \neq \emptyset$, $m = 3$, $s = 0$ and $k \leq 1$. If $k = 1$, all three original missing far heads are exactly the terminal partners, the unique exceptional partner is among them, and the other two already consume the bound $|L| \leq 2$. Their original induced graph has a directed cycle, hence a triangle. The exceptional vertex receives an original arc from a nonexceptional partner on this triangle. The shared head bound puts it in L too, a contradiction. Thus $k = 0$; now $g + m \leq t \leq 4$ gives $g = 1, t = 4$. The residual identity gives the stated size of P . The partition gives $N^{--} = \partial P$, and $\mathcal{F} \subseteq L$, $|L| \leq 4$ gives $L = \mathcal{F}$. Here $\hat{D} = D$ and $|E| = d^{++}(f) + 4 = \mu(f) + |\partial^2 P|$; combine this with $\eta = \mu - 1 + |\partial P| - d^{++} = 4$ to obtain the final equality. $\square$

This threshold does not apply to strictly negative completed gaps. At strong Good equality the supplementary original path theorem gives a finer statement: if $\eta(f) = 2$, every terminal partner is an original nongood second vertex; if $\eta(f) = 3$, there is at most one terminal far partner and it receives an original arc from such a second partner. The unique far partner at residual three may be exceptional. These conclusions leave both the all-second case and the strictly positive Good-slack case unresolved.

6.3 Actual failure sources and the residual margin

For a missing direction $a \rightarrow b$, its original failure set is $\{x : x \rightarrow a, b \in U_D(x)\}$. Let $\mathcal{A}$ be the vertices occurring in the failure set of a direction on a bad missing pair. Opposite directional witnesses are nonadjacent. The cross-witness graph joins every such pair of witnesses. A missing pair is flexible if it is bad or convenient in both directions.

Theorem 6.3 (One uncontrolled actual source). *In an arbitrary oriented graph, suppose some vertex u has the following properties: every actual source in $\mathcal{A} \setminus \{u\}$ has $t \leq 1$, and every cross-witness pair between two such sources is flexible. Then the graph has a Seymour vertex. In MIN, at least two distinct actual failure sources have residual at least two. In particular every positive residual allocation $(m, 1^k)$ is excluded for $m \geq 2$ and finite k .*

Proof. The complete argument is Theorem S27.1. Its first optimization uses all canonical protection-reference completions, maximizing the forward score and then the position of u . A unit-budget feed is balanced. An opposite witness other than u makes a flexible incoming arc backward after sedimentation, whereas witness u moves u strictly later. Thus the selected feed must be u , which therefore has no original protection successor. A separate fixed completion avoiding bad failures at u , followed by a median maximizing its position, yields the second contradiction. The fixed-cover argument is proved independently in Theorem S26.1.

In MIN, an actual source has positive residual; residual one has $t \leq 1$, and the unit-unit opposite-witness pair is bad. If at most one actual source had residual at least two, the general result would apply. The allocation consequence follows even when inactive residuals are unrestricted. $\square$

Theorem 6.4 (Parity-sensitive missing-pair margin). *Every MIN graph satisfies $R = 2M - n \geq 7$, with $R \geq 8$ at even order. Every arc-set-minimal all-deficient graph, whether strong or not, satisfies*

$$M \geq \lceil n/2 \rceil + 3 + \mathbf{1}_{\{n \text{ even}\}}.$$

If o vertices lie outside a sink strong component, the right side can be increased by o . Equality in the uniform bound implies strong connectivity.

Proof. The supplementary two-support theorem gives at least three positive residual vertices (Corollary S12.2). Theorem 6.3 requires two residuals at least two. Hence, at total residual at most six, the only possible positive partitions are

$$(2, 2, 1), \quad (3, 2, 1), \quad (2, 2, 2), \quad (2, 2, 1, 1).$$

The first two are excluded by the general $(m, 2, 1)$ theorem (Theorem S28.7); the last two by Theorems S29.7 and S30.4. These are complete graph arguments, not an enumeration of possible orders. Thus $R \geq 7$; parity gives $R \geq 8$ for even n .

For a sink component of order k with $o = n - k$, the inherited minimality and the outside missing-pair count in Theorem S6.1 give

$$M(D) \geq \lceil k/2 \rceil + 3 + \mathbf{1}_{\{k \text{ even}\}} + \lceil o(k-2)/2 \rceil.$$

An all-deficient graph has minimum outdegree at least two, so $k \geq 5$. Using $k-2 \geq 3$ and checking the two parities yields at least $\lceil n/2 \rceil + o + 3 + \mathbf{1}_{\{n \text{ even}\}}$. If $o > 0$ this is strictly stronger than the uniform bound. $\square$

Thus at $n = 2\delta + 3$ one has $M \geq \delta + 5$, and at $n = 2\delta + 4$ one has $M \geq \delta + 6$. The remaining residual-seven partitions after these filters are

$$(3, 3, 1), \quad (3, 2, 2), \quad (3, 2, 1, 1), \quad (2, 2, 2, 1), \quad (2, 2, 1, 1, 1).$$

Their weak-closure median feeds must have strictly negative completed gap, by Theorem 6.2. None of these five partitions is excluded by that observation alone.

6.4 An additional global constraint on a near-whole feed

Suppose a weak-closure feed f has one original missing partner r . Terminal whole-feed exclusion leaves r still missing. Absence of a weak $r \rightarrow f$ step supplies a current predecessor p of r not pointing to f . Since f is adjacent to every other vertex, $f \rightarrow p$; exact feed neighbourhood invariance gives $r \in N_D^+(f)$. If also $P_D(f)$ is empty, each unprotected original far head y must receive an original arc from r , the only possible missing-partner witness against protection. Put $O = N_D^+(f)$, $d = d_D^+(f)$, $g = g_D(f)$ and $t = t_D(f)$. Then $\mathcal{F}_D(f) \subseteq \partial_D^2 O$. Every member of O reaches f originally within two steps, so $f \in \partial_D^2 O$ as well. Therefore

$$t + 1 \leq |\partial_D^2 O| < |\partial_D O| = d - g, \quad d \geq g + t + 2, \quad n \geq g + 3t + 5.$$

This uses the global original boundary inequality. It is stronger than the constraints obtained by deleting only arcs with tail f , but still leaves unbounded parameter ranges.

7 Exact certificates and reproducibility

The computational component proves a finite statement about cores. It does not enumerate all original graphs covered by Theorem 3.8; their total order is unbounded. The mathematical transfer theorem passes from the finite certificates to those graphs.

7.1 Enumeration and the verification boundary

For a labelled core on k vertices, unordered pairs are listed in lexicographic order. A ternary digit encodes absence, the lower-to-higher direction, or its reverse, with the first pair at the least significant digit. This enumerates all $3^{\binom{k}{2}}$ labelled oriented cores. For each core H , the certificate is a nonzero nonnegative integer vector w . For every set P with a common outneighbour, the verifier checks

$$\sum_{i \notin P: N_H^+(i) \cap P \neq \emptyset} w_i \geq \sum_{i \in P} w_i.$$

Dividing by $\sum_i w_i$ gives the normalized vector used in the proof.

Table 1: Complete incoming-pattern certificate verification.

Core order	Labelled cores	Nonempty inequalities	Largest weight
1	1	0	1
2	3	2	1
3	27	54	1
4	729	3 078	1
5	59 049	459 270	3
Total	59 809	462 404	3

Including one empty pattern for each core gives 522 213 checks. At order five, 59 009 certificates are uniform on their support; the other forty use weights from $\{0, 1, 3\}$. All forty corresponding cores are tournaments, also covered by the analytic density argument. An exact rational solver discovered these forty vectors, but all stored vectors are checked directly with integer arithmetic. Solver correctness is not part of the verifier’s trusted boundary.

The independently organized replays reconstruct adjacency and external predecessor sets from the ternary code. One uses Boolean matrices rather than the generator’s incoming-bitmask unions. They check coverage, vector length, nonnegativity, nonzero mass, and every permitted pattern. Integer sums in the supplied tables are at most nine; the principal replay also checks that all sums are safe integers.

```

1 for k = 1,...,5:
2   assert table[k].length == 3^(k*(k-1)/2)
3   for code = 0,...,table[k].length-1:
4     H = decode_oriented_graph(k, code)
5     w = table[k][code]
6     assert w is a nonzero vector of nonnegative integers
7     for P subset of V(H):
8       if some j receives an arc from every vertex of P:
9         R = {i outside P: i points to some vertex of P}
10        assert sum(w[i] for i in R) >= sum(w[i] for i in P)

```

Listing 1: Independent certificate verification, pseudocode.

The supplied verification directory can be checked with standard Node.js, without a solver, network access or third-party packages:

```

1 node verify_small_core_certificates.js
2 node verify_core_pattern_certificates_independent.js

```

The SHA-256 checksum of the order-five table is

c9fe139b76a3866a7bf17e408d12f344823fa
1a54fca306fd656139e601bfd5a

The line break is typographical. Reproducing the generator is unnecessary to validate a supplied certificate.

7.2 Controls for the completion arguments

Finite graph checks supplement the closure proofs but do not establish their universal quantifiers. The following controls test their scope. In an adjacency mask, bit j of the i th integer means $i \rightarrow j$.

The six-vertex graph $(4, 4, 24, 32, 32, 3)$ has weak closure $(6, 4, 24, 48, 32, 3)$ and a maximizing completion $(30, 28, 56, 48, 32, 3)$. Its unique median is $(0, 1, 2, 3, 4, 5)$. At feed 5, the original gap is one, the completed gap is minus one, $t_D = 2$, and the terminal missing degree is one. Thus the general $+1$ surplus is attained in the negative-gap branch. Good slack is one, so equality sedimentation cannot move the order.

The eleven-vertex original masks

$$(6, 8, 8, 17, 1793, 80, 144, 48, 33, 65, 129)$$

give terminal partners $\{5, 6, 7\}$ at feed 0, with original cycle $5 \rightarrow 6 \rightarrow 7 \rightarrow 5$. Exact subset dynamic programming verifies partial median score 23 and maximizing tournament score 49 for the reference $(8, 9, 10, 5, 6, 7, 1, 2, 3, 4, 0)$. Here $g_D(0) = 1$, $g_T(0) = 0$ and $t_D(0) = 4$, attaining the zero-gap $+3$ bound. Both controls contain Seymour vertices and are not counterexamples to the conjecture or to the MIN residual estimates.

A ten-vertex control also satisfies all outgoing-star deletion conditions at one feed, with local residual two, yet has stable negative completed gap. Its original first boundary and external second boundary both have size three, violating the global boundary lemma. This distinguishes single-feed criticality from full original arc-set minimality. Its explicit data and the exact-score programs are supplied with the supplementary records.

7.3 Other computations and their scope

Copying and closure programs test identities on arbitrary graphs, including several changed rows and shared heads. They do not simulate a hypothetical all-deficient minimal graph. Searches that fail to find a graph or a difficult branch are not proofs of nonexistence. Earlier minimum-degree-seven branch logs are archived as local computational records and are not used to infer a complete minimum-degree theorem. The exact five-core theorem has the exhaustive finite coverage specified above, independently of those searches.

8 Further consequences and unresolved steps

The incoming-pattern formulation isolates a precise global question: does every finite oriented core admit a pattern certificate? Theorem 3.10 shows that universal feasibility would settle Seymour's conjecture, while an infeasible core would yield a concrete counterexample. The present tables establish only the core-order-five range. Iterated incoming-core reduction removes repeated or contained incoming sets from the remaining certificate problem, but does not prove feasibility on all larger terminal cores.

The copying and completion methods impose different constraints. In a minimum-order counterexample the maximal gap-one classes cover every target; a further lexicographic choice

makes incoming classes true independent modules. A closure feed can nevertheless remain originally deficient because it gains several second neighbours. The zero-gap branch has an original partner cycle and residual at least four. The negative-gap branch can have strict Good slack or an original nongood second partner. We have not proved that the original unit-class cover eliminates all these possibilities.

Two supplementary results delimit other approaches. Cyclic three-copy amplification sends order n and minimum outdegree δ to

$$n' = 3n, \quad \delta' = n + \delta, \quad n' - 2\delta' = n - 2\delta,$$

while preserving vertex gaps, missing degrees and local residuals. A fixed near-half minimum-degree theorem for all oriented graphs would therefore imply the conjecture. Amplification does not preserve arc-set minimality, so a minimal residual bound cannot simply be transferred to the amplified dense graph.

For a fixed protection-preserving tournament completion T , set $\Delta = g_T - g_D$, and let N_T be the signed second-minus-first matrix. The exact correction budget has dual form

$$\max_{\substack{\lambda \geq 0, \mathbf{1}^T \lambda = 1 \\ N_T^T \lambda \geq 0}} \lambda^T \Delta = \min_{w \geq 0} \max_x (\Delta_x + (N_T w)_x).$$

An explicit six-class clone family has optimum $-k/3$ for every canonical reference completion, even after strong dense amplification. These graphs already have Seymour vertices. They refute an unconditional completion-budget claim, not the conjecture. The supplement supplies a common integer dual certificate for the whole family. A normalized correction strictly greater than -1 would suffice for an all-deficient graph; its universal existence remains unproved.

The specialized $n = 2\delta + 3$ results in the supplement include exact deletion identities, minimum-degree predecessor estimates, capacity and heavy-set constraints, and conditional degree thresholds. Their capacity variable σ is distinct from η . Earlier finite pure-unit bounds and one-heavy exclusions are recorded as consequences of the unbounded source theorem rather than as independent contributions.

Verification and attribution. The arguments and programs underwent independent internal reconstruction and replay. This is not a proof-assistant certificate or an external referee report. AI tools assisted research, drafting and implementation; human authorship, review and approval are required before submission. Literature comparison establishes the cited inputs and their scope, but does not certify priority for every auxiliary observation. The source, certificate tables, dependency inventory and verification records support further examination. No complete proof of the general conjecture or actual counterexample is asserted.

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